Numerical methods for the simulation of air flow past a mountain

Master’s thesis defence

Bc. Kamil Belán

Supervisor: doc. RNDr. Michal Pavelka, Ph.D.

Consultant: doc. RNDr. Petr Šácha, Ph.D.

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Presentation outline

Problem settings

Mathematical description

Numerical solution

Results

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Physical phenomena

- nontrivial topography $\Longrightarrow$ *multiscale atmospheric phenomena*
- goal: capture **internal gravity waves (IGWs)**
- **IGWs influence**

Mountain waves. Source: Marc Thunis

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  - transport and mixing

Mountain waves. Source: Marc Thunis

Physical phenomena

- nontrivial topography $\Longrightarrow$ *multiscale atmospheric phenomena*
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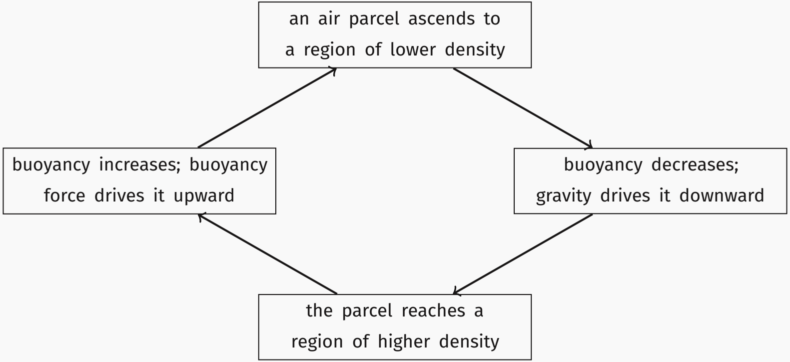
- **IGWs influence**
  - (clear air) turbulence
  - transport and mixing
  - downhill windstorms

Mountain waves. Source: Marc Thunis

Internal gravity wave prototype

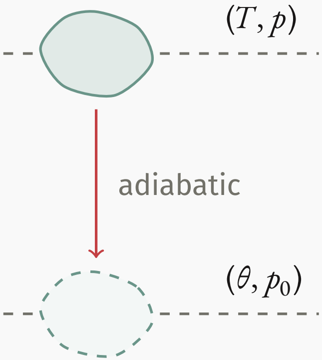
- *air density* and *atmospheric pressure* are stratified
- a displaced air parcel is pushed back by buoyancy & gravity $\Longrightarrow$ **internal gravity wave**



- initial displacement caused by hitting the mountain profile

Potential temperature θ

- thermodynamic temperature a parcel would have if moved **adiabatically** from (T, p) to (θ, p_0)
- conserved under *isentropic flows*
- **stratification stability condition:** $\partial_z \theta > 0$.



$p_0 = 1013.25$ hPa

Potential temperature expression

$$\theta = T \left(\frac{p_0}{p} \right)^{\frac{R}{c_p}}.$$

Mathematical description

Governing equations

Compressible isentropic Euler equations

$$\rho \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla p + \rho \mathbf{g}, \qquad \text{momentum balance}$$

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} + \rho \left(\nabla \cdot \mathbf{v}\right) = 0, \qquad \text{mass conservation}$$

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = 0, \qquad \text{isentropic evolution, } \theta = T\left(\frac{p_0}{p}\right)^{\frac{R}{c_p}}$$

$$p\left(\rho, T\right) = \rho R T. \qquad \text{thermal equation of state of an ideal gas}$$

Initial conditions

Isothermal hydrostatic balance, $T_0 = 250\,\mathrm{K}$

$$
\begin{aligned}
\mathbf{v}\,(0,\mathbf{x}) &= \mathbf{0}, \\
\rho\,(0,\mathbf{x}) &= \bar{\rho}(z) := \rho_0 \exp\left(-z\frac{g}{RT_0}\right), \\
\theta\,(0,\mathbf{x}) &= \bar{\theta}(z) := T_0 \exp\left(z\frac{R}{c_p}\frac{g}{RT_0}\right), \\
p\,(0,\mathbf{x}) &= \bar{p}(z) := \rho_0 R T_0 \exp\left(-z\frac{g}{RT_0}\right) \qquad \Leftarrow \text{equation of state.}
\end{aligned}
$$

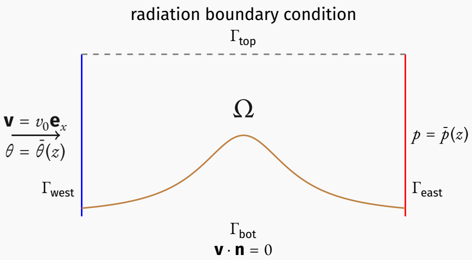
Domain & boundary conditions

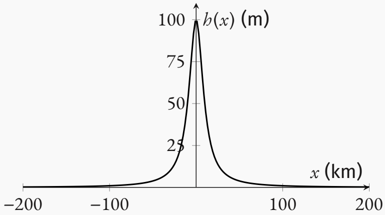
$\Omega \subset \mathbb{R}^2, \partial\Omega = \Gamma_{\text{west}} \cup \Gamma_{\text{east}} \cup \Gamma_{\text{bot}} \cup \Gamma_{\text{top}}$

Imposed boundary conditions

$\mathbf{v} = v_0 \mathbf{e}_x$ on Γ_{west},	subsonic inflow: $v_0 < c$
$\theta = \bar{\theta}(z)$, on Γ_{west},	hydrostatic profile
$\mathbf{v} \cdot \mathbf{n} = 0$ on Γ_{bot},	no-penetration
$p = \bar{p}(z)$ on Γ_{east},	subsonic outflow

together with the *radiation boundary condition* on the *artificial* boundary Γ_{top}.





Numerical solution

Smoothed Particle Hydrodynamics

- fluid is partitioned into *abstract material points* $\{\mathbf{x}_a\}$, so called **particles**
- equations are discretised using an **interpolation kernel** $w_h(\mathbf{x})$, $h \propto \operatorname{diam} \operatorname{supp} w_h$

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$$\Phi(t,\mathbf{x}) = \Phi(t,\mathbf{x}) \star \delta(\mathbf{x}) = \int_\Omega \Phi(t,\mathbf{x}')\delta(\mathbf{x}-\mathbf{x}')\,\mathrm{d}\mathbf{x}' \implies$$

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particle approximation

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- the resulting system of *ODEs* is solved numerically using a **symplectic integrator**

Why SPH?

- **conservative** properties, accessible **trajectories**

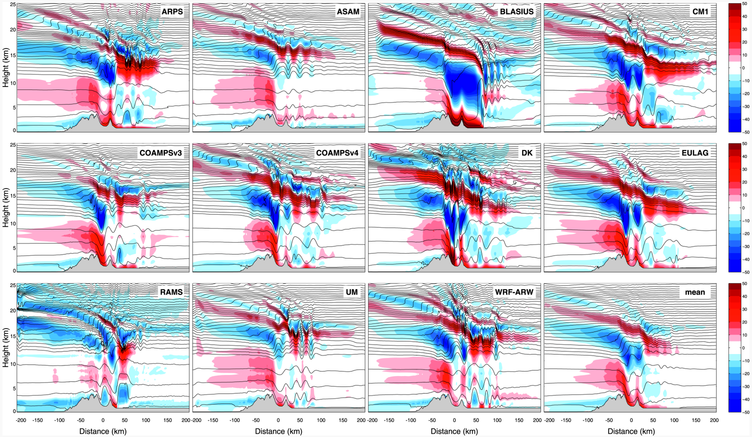

Figure: Reference results: vertical component of the velocity field Monthly Weather Review 139, 9; 10.1175/MWR-D-10-05042.1

$p - \theta$ formulation

Standard SPH

$$\frac{\mathrm{d}\rho_a}{\mathrm{d}t} = \sum_b m_b w'_{ab} \mathbf{v}_{ab} \cdot \mathbf{e}_{ab},$$

$$\frac{\mathrm{d}\mathbf{v}_a}{\mathrm{d}t} = -\sum_b m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2} \right) w'_{ab} \mathbf{e}_{ab} + \mathbf{g},$$

$$\theta = T_a \left(p_0 / p_a \right)^{\frac{R}{c_p}} \approx \text{const},$$

- poor conservation properties with adaptive h,
- θ has to be computed at extra cost.

Novel $p - \theta$ formulation

$$p_a = \left(\sum_b \frac{m_b}{\tilde{\theta}_0} \theta_b w_{ab} \left(h_a \right) \right)^{\gamma},$$

$$\frac{\mathrm{d}\mathbf{v}_a}{\mathrm{d}t} = -\sum_b m_b \frac{\theta_a \theta_b}{\tilde{\theta}_0^2} \left(\frac{f_a p_a}{p_a^{2/\gamma}} w'_{ab} \left(h_a \right) + \frac{f_b p_b}{p_b^{2/\gamma}} w'_{ab} \left(h_b \right) \right) \mathbf{e}_{ab} + \mathbf{g},$$

$$\theta_a = \text{const}$$

- naturally guarantees conservation with adaptive h via the ∇h terms f_a,
- uses the primary variable θ directly.

Well-balanced SPH formulations

- hydrostatic $\bar{\rho}, \bar{p}$ solve

$$\cancel{\bar{\rho}\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}} + \nabla\bar{p} - \bar{\rho}\mathbf{g} = \mathbf{0},$$

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- general numerical approximations $\bar{\rho}_h, \bar{p}_h, \nabla_h$
$$\nabla_h \bar{p}_h - \bar{\rho}_h \mathbf{g} \neq \mathbf{0} \implies \textbf{spurious force}$$

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Hydrostatic balance does not generally hold on the discrete level!

- **well-balanced scheme**: stationary states preserved **after discretisation**

Well-balancing: substituting for gravity

$$\mathbf{g} = \frac{1}{\bar{\rho}}\nabla\bar{p} \implies \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\frac{1}{\rho}\nabla p + \frac{1}{\bar{\rho}}\nabla\bar{p}$$

- identical discretisation of gradients $\implies$ identical truncation errors $\implies$ they **subtract**

Results

The failure of standard SPH: velocity field

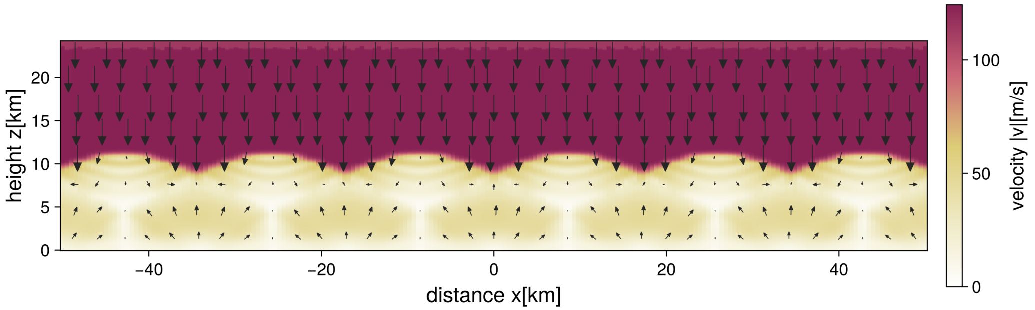


Figure: Nonzero velocity field computed by standard SPH - a failure of hydrostatic balance.

The success of well-balanced $p - \theta$: velocity norms

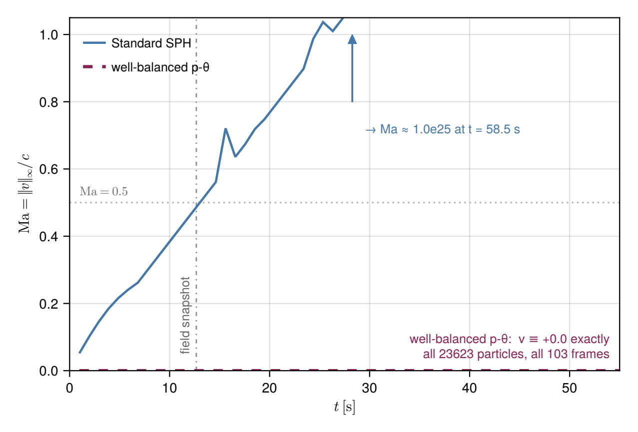
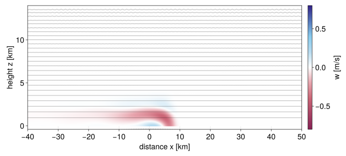
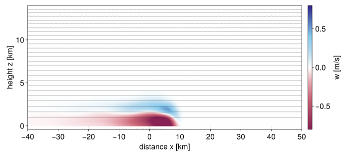


Figure: Comparison of standard SPH and well-balanced $p - \theta$. $p - \theta$ preserves hydrostatic balance exactly.

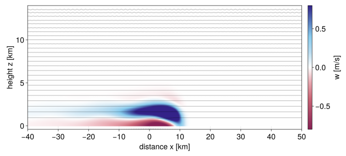
Partial success: emergence of a wave



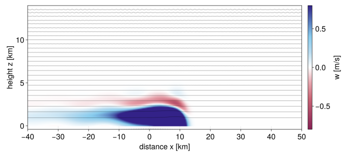
(a) t = 360 s



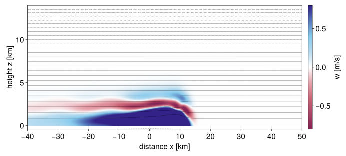
(b) t = 420 s



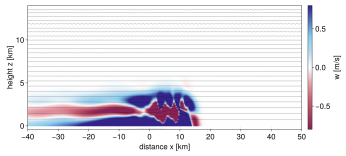
(c) t = 480 s



(d) t = 540 s



(e) t = 600 s



(f) t = 660 s

Tensile instability: clumping

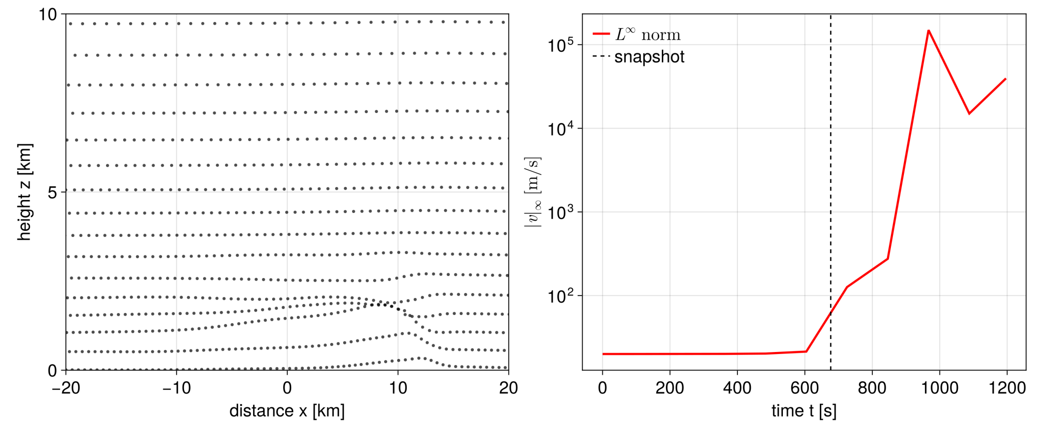


Figure: $t = 690$ s

Tensile instability: explosion

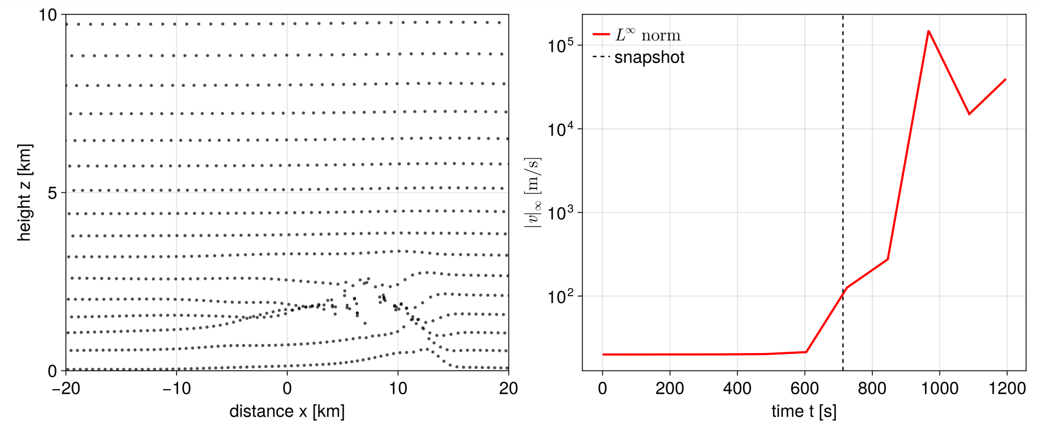


Figure: $t = 710$ s

Tensile instability: breakdown

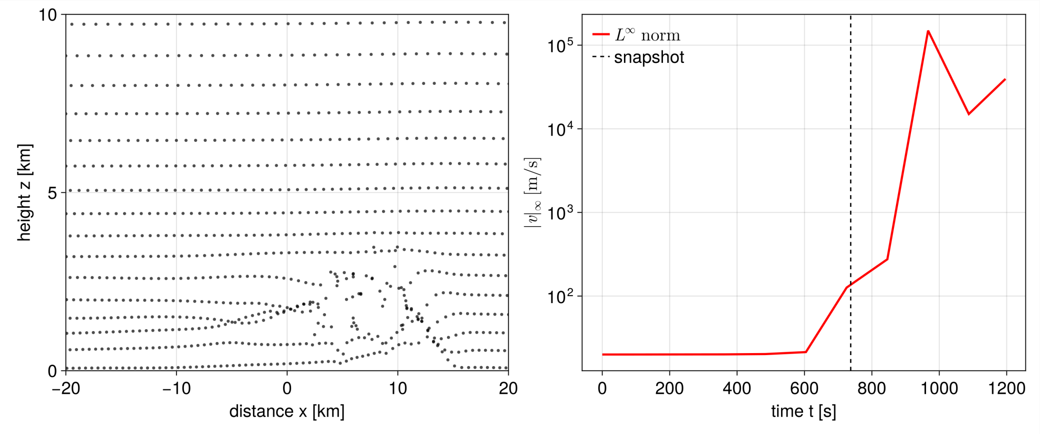
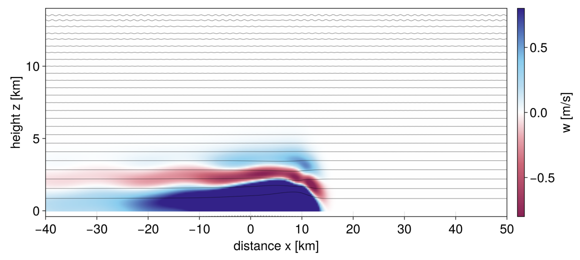


Figure: $t = 720$ s

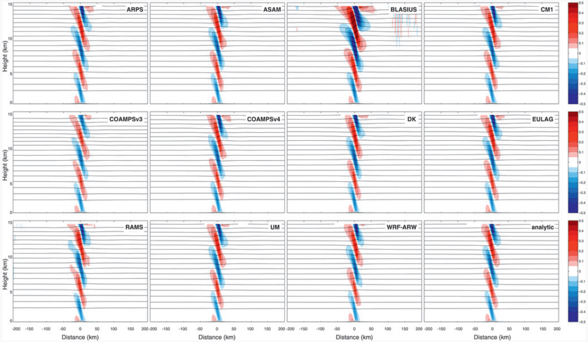
Summary

- flows around a topography induce internal gravity waves
- IGWs play a significant role in mesoscale atmospheric dynamics
- large scale hydrostatic balance is problematic for standard SPH
- well-balanced schemes with adaptive smoothing lengths must be used for non-stationary flows



Future plans

- resolve tensile instability by δ–SPH regularization
- implement other boundary conditions, include real topography
- compare the results with the analytic solution
- compare the results with other models & methods.



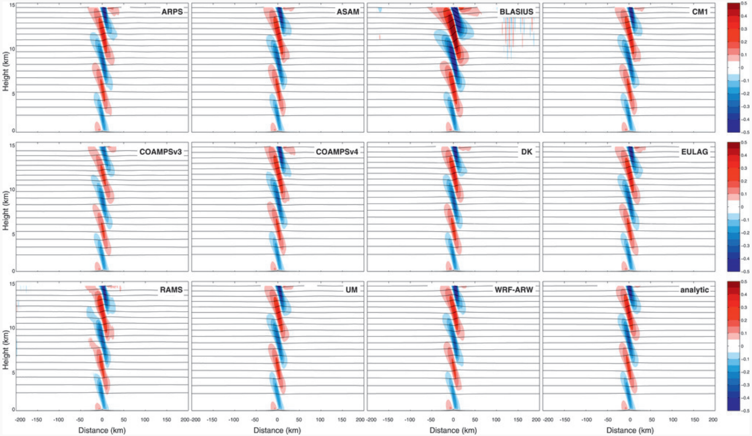
Reference results. Monthly Weather Review 139, 9; 10.1175/MWR-D-10-05042.1

Reakce na otázky z posudků

Oponent, otázka 1

> *„Vztah (2.4), uvažovala/testovala se i jiná volba okrajových podmínky na vstupu a výstupu, viz komentář v práci na str. 27?“*

Oponent, otázka 1: odpověď



Obrázek: Reference results. Monthly Weather Review 139, 9; 10.1175/MWR-D-10-05042.1

Oponent, otázka 2

> *„Numerické experimenty v Kapitole 6 potvrzují schopnost metody zachovávat stacionární stavy. Částečně však postrádám další numerické verifikace, např. vyšetřování přesnosti či stability metody. Byly prováděny též experimenty v této oblasti?“*

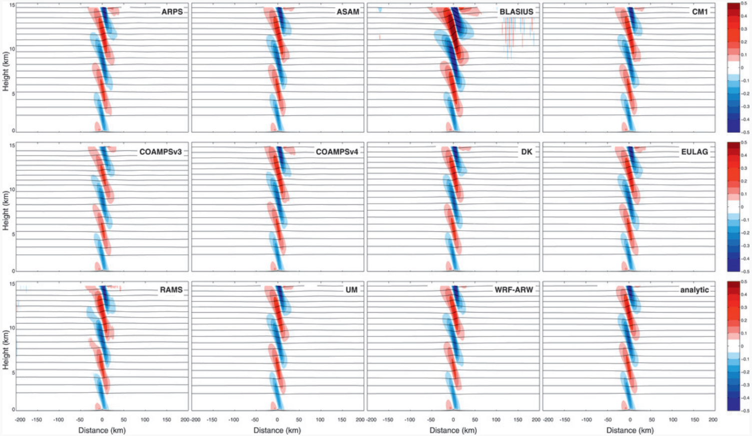
Oponent, otázka 3

> *„Na základě zkušeností s řešením práce, jaký je potenciál částicových metod ve srovnání se „sítovými metodami“ (FEM, FVM, DG) pro řešení komplikovanějších 3D úloh?“*

Vedoucí, otázka 1

> *„Dá se pozorovaná vertikální vlna nazvat gravitační vlnou v meteorologickém kon-
> textu?“*

Vedoucí, otázka 1: odpověď



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Vedoucí, otázka 2

„Dal by se proměnlivý kernel začlenit do hamiltonovské formulace?"

Odpověď

Obecná evoluce

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \partial_t F + \{F, E\}, \tag{1}$$

SPH závorka

$$\{F(\mathbf{x}_a, \mathbf{M}_a), G(\mathbf{x}_a, \mathbf{M}_a)\} = \sum_a \left(\frac{\partial F}{\partial x_a^i} \frac{\partial G}{\partial M_{a,i}} - \frac{\partial F}{\partial M_{a,i}} \frac{\partial G}{\partial x_a^i} \right), \tag{2}$$

SPH energie

$$E = \sum_a \left(\frac{|\mathbf{M}_a|^2}{2m_a} + \frac{m_a}{\rho_a} e(\rho_a, \eta_a) \right), \tag{3}$$

Vedoucí, otázka 2

Odpověď (pokračování)

Evoluce rychlosti

$$\frac{\mathrm{d}\mathbf{v}_a}{\mathrm{d}t} = \frac{1}{m_a}\frac{\mathrm{d}\mathbf{M}_a}{\mathrm{d}t} = -\frac{1}{m_a}\nabla_a E, \tag{4}$$

SPH energie

$$E = \sum_a \left(\frac{|\mathbf{M}_a|^2}{2m_a} + \frac{m_a}{\rho_a} e\left(\rho_a, \eta_a\right) \right), \tag{5}$$

Hustota

$$\rho_a = \sum_b m_b w_{ab}\left(h_b\right) \tag{6}$$

Vedoucí, otázka 3

> *„Jak obtížné by bylo vyvinout well-balanced implementaci SPH s δ–SPH regularizací?“*